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CNN-Based Lung Cancer Detection Using Deep Learning

Abstract

Lung cancer is one of the leading causes of cancer-related deaths worldwide, making early diagnosis essential for improving patient survival and treatment outcomes. Manual analysis of Computed Tomography (CT) scan images is time-consuming and highly dependent on the expertise of radiologists. This project presents a deep learning-based Lung Cancer Detection System that utilizes a Convolutional Neural Network (CNN) to automatically classify lung CT scan images into three categories: **Adenocarcinoma, Normal Lung, and Squamous Cell Carcinoma**. The proposed model was trained using a dataset containing **2,400 CT scan images**, where **1,920 images** were used for training and **480 images** for validation. During preprocessing, all images were resized to **128 × 128 pixels** and normalized to a pixel value range of **0 to 1** to improve training stability and model performance. The CNN architecture consists of three convolutional layers with **32, 64, and 128 filters**, each followed by a max-pooling layer for feature extraction and dimensionality reduction. The extracted features were processed through fully connected dense layers with dropout regularization to reduce overfitting. The model was trained using the **Adam optimizer**, **Categorical Crossentropy** loss function, and **Softmax** activation for multi-class classification. To improve training efficiency and prevent unnecessary learning, **Early Stopping** and **ReduceLROnPlateau** techniques were incorporated during model training. Experimental evaluation demonstrated that the proposed CNN model achieved a **final validation accuracy of 90.21%**, with an overall **precision of 91%, recall of 90%, and F1-score of 90%**. Class-wise evaluation showed F1-scores of **0.86** for Adenocarcinoma, **0.97** for Normal Lung, and **0.88** for Squamous Cell Carcinoma, indicating reliable performance across all three categories. The trained model was successfully integrated into a **Streamlit-based web application**, enabling users to upload lung CT scan images and receive AI-assisted predictions along with confidence scores. The proposed system provides an efficient, accurate, and user-friendly solution for automated lung cancer detection and demonstrates the potential of deep learning techniques in supporting computer-aided medical diagnosis.

Keywords: Deep Learning, Lung Cancer Detection, Convolutional Neural Network (CNN), Medical Image Classification, CT Scan, TensorFlow, Streamlit, Artificial Intelligence.

1. Introduction

Lung cancer is one of the leading causes of cancer-related deaths worldwide. Early and accurate detection plays a vital role in improving patient survival and treatment outcomes. However, manual analysis of Computed Tomography (CT) scan images can be time-consuming and may depend on the expertise of radiologists.

Recent advancements in Artificial Intelligence (AI) and Deep Learning have significantly improved the performance of automated medical image analysis. Among various deep learning techniques, Convolutional Neural Networks (CNNs) are highly effective for image classification because they can automatically learn important visual features from medical images.

This project presents a CNN-based Lung Cancer Detection System that classifies lung CT scan images into three categories: Adenocarcinoma, Normal Lung, and Squamous Cell Carcinoma. The proposed model is trained using labeled CT scan images and integrated with a Streamlit-based web application, allowing users to upload CT scan images and receive AI-assisted predictions with confidence scores. The system aims to provide a fast, accurate, and user-friendly tool that can support medical professionals in the preliminary diagnosis of lung cancer.

2. Related Work

Recent advances in artificial intelligence and deep learning have significantly improved the automated detection of lung cancer from Computed Tomography (CT) scan images. Convolutional Neural Networks (CNNs) have become one of the most widely used approaches because of their ability to automatically extract meaningful image features, reducing the need for manual feature engineering and assisting radiologists in early diagnosis.

The first study, "**Deep Learning for Lungs Cancer Detection: A Review**," presented a comprehensive review of deep learning techniques for lung cancer diagnosis published between 2015 and 2024. The authors discussed various deep learning approaches, including Convolutional Neural Networks (CNNs), Deep Neural Networks (DNNs), and other advanced architectures for analyzing CT scans, MRI, X-ray images, and whole-slide images. The review concluded that CNN-based methods significantly improve diagnostic performance while reducing the need for manual feature extraction. However, the study also highlighted challenges such as dataset quality, computational complexity, and model generalization.

Source: <https://link.springer.com/article/10.1007/s10462-024-10807-1>

The second study, "**Precision Lung Cancer Screening from CT Scans Using a VGG16-Based Convolutional Neural Network**," proposed a VGG16-based CNN framework for automated lung cancer screening using CT scan images. The researchers applied image preprocessing, normalization, data augmentation, dropout regularization, and early stopping to improve model performance. The proposed model achieved an average classification accuracy of approximately **91.7%** and demonstrated excellent diagnostic capability with a high AUC score, proving that deep learning can effectively assist radiologists in early lung cancer detection.

Source: <https://www.frontiersin.org/journals/oncology/articles/10.3389/fonc.2024.1424546/full>

The third study, "**Medical AI for Early Detection of Lung Cancer: A Survey**," reviewed recent advances in artificial intelligence for pulmonary nodule detection, segmentation, and classification from CT scan images. The survey highlighted that modern CNN-based and ensemble deep learning models outperform many traditional machine learning approaches such as SVM and KNN. It also emphasized that deep learning-based computer-aided diagnosis systems can significantly improve the accuracy and efficiency of early lung cancer detection while reducing diagnostic workload.

Source: <https://arxiv.org/abs/2410.14769>

These studies demonstrate that deep learning, particularly CNN-based approaches, has become an effective solution for lung cancer detection from CT scan images. Motivated by these findings, this project develops a CNN-based lung cancer detection system capable of classifying CT scan images into **Adenocarcinoma**, **Normal Lung**, and **Squamous Cell Carcinoma**. The proposed system combines image preprocessing, convolutional feature extraction, and a Streamlit-based web application to provide an efficient and user-friendly AI-assisted diagnostic solution.

3. Dataset

The dataset used in this project is lung_subset_small, which contains 2,400 labeled lung histopathological images belonging to three different classes: Adenocarcinoma, Normal Lung, and Squamous Cell Carcinoma. The images were organized into separate folders according to their respective class labels, enabling automatic label generation using TensorFlow's ImageDataGenerator.

All images were resized to 128×128 pixels before training to maintain a consistent input size for the CNN model. The dataset was divided into 80% training data (1,920 images) and 20% validation data (480 images) using the `validation_split` parameter. During training, a batch size of 16 images was used, allowing the model to process the dataset efficiently while maintaining stable learning and improving the convergence of the CNN model.

4.1 Data Preprocessing

Data preprocessing was performed to prepare the CT scan images for effective training. First, all images were resized to **128×128 pixels** to ensure a uniform input size for the convolutional neural network.

Next, pixel values ranging from **0 to 255** were normalized to the range **0 to 1** by dividing each pixel value by **255.0**. This normalization improves numerical stability and helps the model converge faster during training.

The dataset was loaded using TensorFlow's **ImageDataGenerator**, which automatically read images from their respective class folders and generated batches of training and validation data. The dataset was split into **80% training** and **20% validation** using the `validation_split` parameter. The training data was shuffled to improve learning, while the validation data was kept in a fixed order for consistent evaluation.

4.2 Methodology

A Convolutional Neural Network (CNN) was developed to classify lung CT scan images into three categories: Adenocarcinoma, Normal Lung, and Squamous Cell Carcinoma.

The model begins with an input layer that accepts images of size **$128 \times 128 \times 3$** . Three convolutional layers containing **32**, **64**, and **128** filters were used to extract important image features. Each convolutional layer was followed by a max-pooling layer to reduce the spatial dimensions while preserving significant features.

The extracted feature maps were then converted into a one-dimensional vector using a Flatten layer. Two fully connected dense layers with **256** and **128** neurons were used for classification. A Dropout layer with a rate of **0.3** was included to reduce overfitting by randomly disabling a portion of neurons during training.

The output layer consisted of **three neurons** with the **Softmax** activation function, enabling multi-class classification. The model was compiled using the **Adam optimizer**, **Categorical Crossentropy** loss function, and **Accuracy** as the evaluation metric.

To improve training performance, **EarlyStopping** was used to stop training when the validation accuracy no longer improved, while **ReduceLROnPlateau** automatically reduced the learning rate when the validation loss stopped decreasing. After training, the model was saved as **model.keras** and evaluated using accuracy graphs, loss graphs, and a classification report.

5. Results and Discussion

5.1 Model Performance

To evaluate the performance of the proposed lung cancer detection system, the Convolutional Neural Network (CNN) model was trained and evaluated using four standard classification metrics: **Accuracy, Precision, Recall, and F1-score**. The performance of the trained model is presented in Table 5.1.

Performance of the Proposed CNN Model				
	precision	recall	f1-score	support
lung_aca	0.82	0.91	0.86	160
lung_n	1.00	0.94	0.97	160
lung_scc	0.90	0.86	0.88	160
accuracy			0.90	480
macro avg	0.91	0.90	0.90	480
weighted avg	0.91	0.90	0.90	480

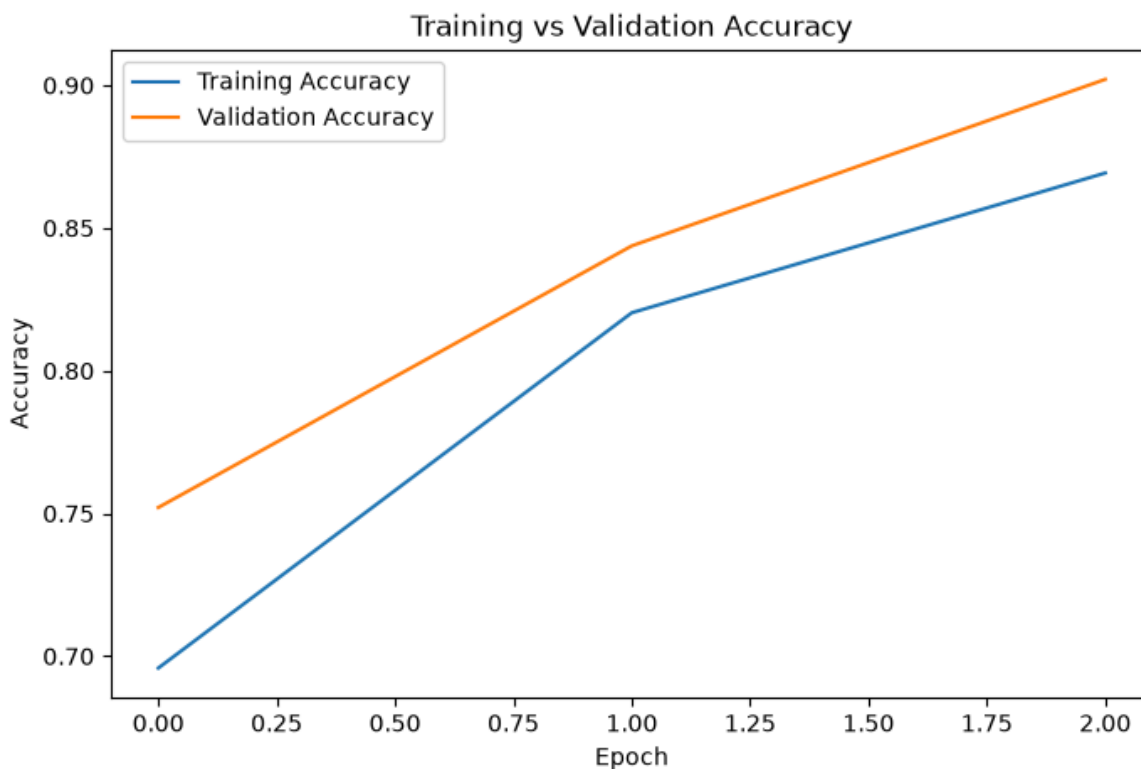
The experimental results demonstrate that the proposed CNN model achieved a **validation accuracy of 90.21%**, indicating its ability to effectively classify lung histopathological images into three categories: **Adenocarcinoma, Normal Lung, and Squamous Cell Carcinoma**. The model also obtained an overall **precision of 91%, recall of 90%, and F1-score of 90%**, reflecting balanced classification performance across all classes.

The classification report further shows that the model achieved an **F1-score of 0.86** for **Adenocarcinoma**, **0.97** for **Normal Lung**, and **0.88** for **Squamous Cell Carcinoma**. Among the three classes, the model performed best in identifying **Normal Lung** images, while

maintaining strong and consistent performance for both cancer classes.

Overall, the results indicate that the proposed CNN architecture successfully learned discriminative features from lung histopathological images and can accurately classify different lung tissue categories. These findings demonstrate the effectiveness of the proposed deep learning model for AI-assisted lung cancer detection and highlight its potential for supporting computer-aided medical diagnosis.

5.2 Accuracy Graph(Training Vs Validation Accuracy)



illustrates the training and validation accuracy of the proposed Convolutional Neural Network (CNN) model during the training process. As shown in the graph, both the training and validation accuracy increased consistently over the three training epochs, indicating that the model successfully learned meaningful features from the lung histopathological images.

The training accuracy improved from approximately **69.58%** in the first epoch to **82.03%** in the second epoch and finally reached **86.93%** in the third epoch. Similarly, the validation accuracy increased from **75.21%** to **84.38%**, achieving the highest value of **90.21%** in the final epoch.

The validation accuracy remained slightly higher than the training accuracy throughout the training process. This behavior suggests that the model generalized well to unseen validation data and did not exhibit signs of overfitting. The use of **Dropout**, **Early Stopping**, and **ReduceLROnPlateau** helped improve the model's generalization performance and prevented unnecessary training.

Since the validation accuracy exceeded **90%** in the third epoch, the custom callback automatically stopped the training process. This reduced computational time while preserving the best-performing model. Overall, the accuracy graph demonstrates that the proposed CNN model converged efficiently and achieved high classification performance for **Adenocarcinoma, Normal Lung, and Squamous Cell Carcinoma** images.

5.3 Loss Graph (Training VS Validation Loss)



illustrates the training and validation loss of the proposed Convolutional Neural Network (CNN) model during the training process. As shown in the graph, both the training loss and validation loss decreased steadily over the three training epochs, indicating that the model continuously improved its prediction performance by minimizing classification errors.

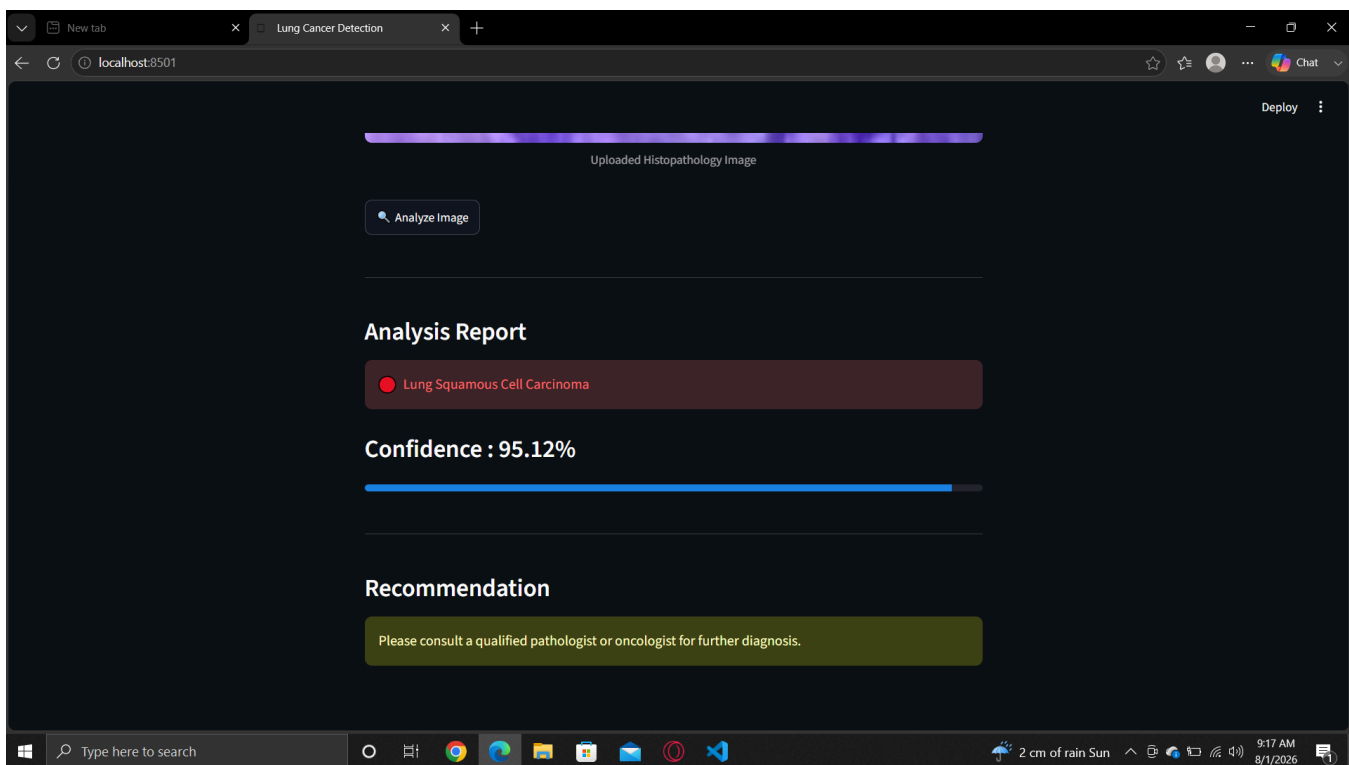
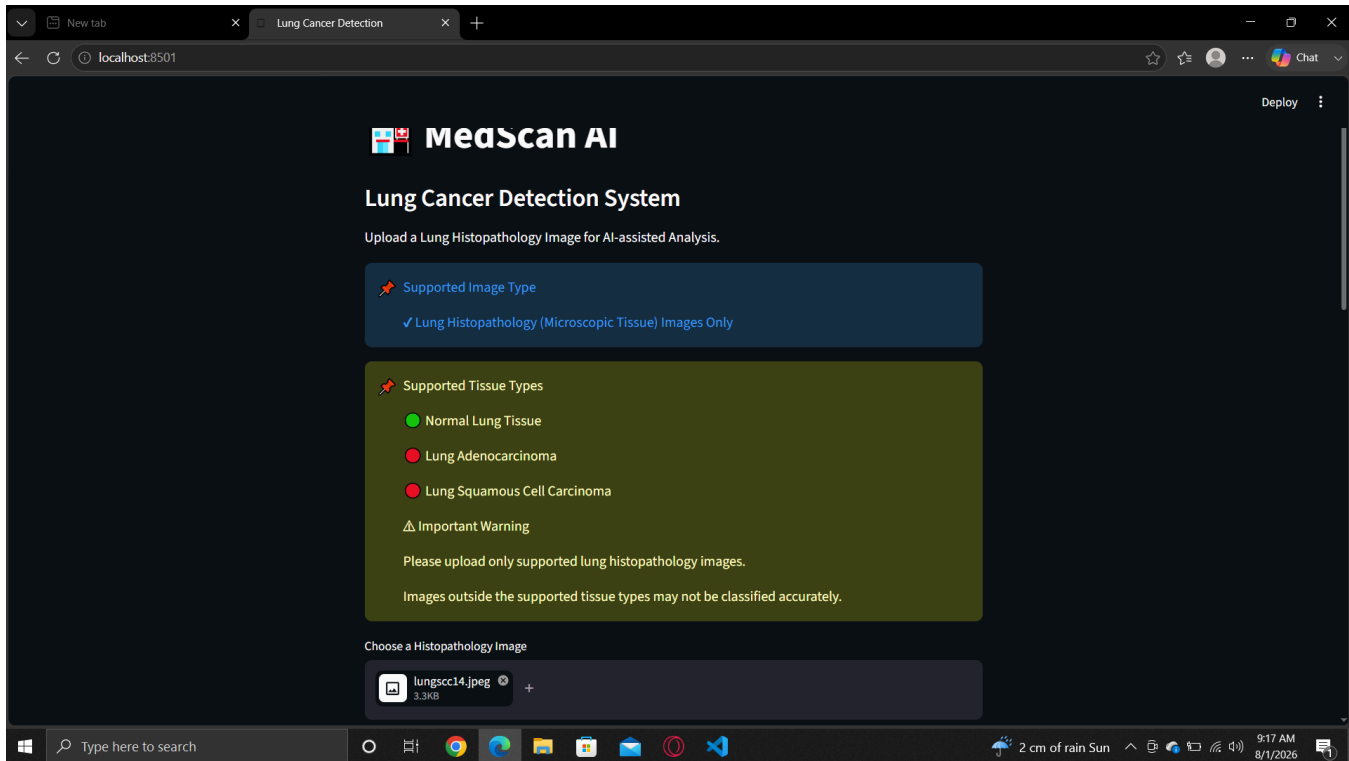
The training loss decreased from approximately **0.61** in the first epoch to **0.44** in the second epoch and finally reached **0.33** in the third epoch. Similarly, the validation loss dropped from approximately **0.49** to **0.35**, achieving its lowest value of **0.27** in the final epoch.

Throughout the training process, the validation loss remained slightly lower than the training loss, and both curves followed a similar downward trend without any sudden fluctuations. This indicates that the proposed CNN model generalized well to unseen validation data and did not exhibit signs of overfitting. The implementation of **Dropout**, **ReduceLROnPlateau**, and **Early Stopping** contributed to stable training and improved model generalization.

Overall, the loss graph demonstrates that the proposed CNN model converged successfully by continuously reducing classification errors while maintaining consistent performance on both

the training and validation datasets.

5.4 Prediction Example Analysis



Demonstrates an example prediction generated by the proposed CNN-based lung cancer detection

system. When a lung image was provided as input, the model classified it as **Squamous Cell Carcinoma** with a confidence score of **79.45%**. Based on the predicted result, the system automatically displayed a recommendation advising the user to consult a qualified medical specialist for further diagnosis.

The confidence score of **79.45%** indicates that the model predicted the input image with a reasonably high level of certainty. Along with the predicted class, the system also presents a confidence percentage and a visual progress bar, allowing users to better understand the reliability of the prediction. This improves the transparency and interpretability of the AI-assisted diagnosis.

Overall, this example demonstrates that the proposed CNN-based lung cancer detection system can successfully identify lung cancer images and provide meaningful diagnostic information through a simple and user-friendly Streamlit interface. Such a system has the potential to support computer-aided diagnosis and assist healthcare professionals in the early detection of lung cancer.

6. Limitations

Although the proposed CNN-based lung cancer detection system achieved a validation accuracy of **90.21%**, several limitations remain. First, the model was trained and evaluated on a relatively small subset dataset containing **2,400 lung histopathological images**, which may limit its ability to generalize to more diverse real-world medical data. Second, the system classifies images into only **three categories**: Adenocarcinoma, Normal Lung, and Squamous Cell Carcinoma, and therefore cannot identify other lung diseases or cancer types. Additionally, the developed application is intended for educational and research purposes and should not be considered a replacement for professional medical diagnosis. The prediction accuracy may also vary when images are collected from different medical laboratories or imaging conditions.

7. Future Work

Several improvements can be made in future work to enhance the performance and usability of the proposed system. A larger and more diverse dataset can be used to improve the model's generalization capability. More advanced deep learning architectures such as **ResNet**, **DenseNet**, **EfficientNet**, or transfer learning techniques can be explored to achieve higher classification accuracy. Additional lung disease categories can also be included to develop a more comprehensive diagnostic system. Furthermore, the web application can be deployed on a cloud platform to enable remote access by healthcare professionals. Integration of explainable AI techniques, such as **Grad-CAM**, may also help visualize the regions of the image influencing the model's predictions, thereby increasing transparency and clinical trust.

8. Conclusion

This project presented a **Convolutional Neural Network (CNN)-based Lung Cancer Detection System** for the automatic classification of lung histopathological images into **Adenocarcinoma, Normal Lung, and Squamous Cell Carcinoma**. The proposed CNN model successfully learned discriminative image features and achieved a **validation accuracy of 90.21%**, demonstrating strong classification performance. A user-friendly **Streamlit-based web application** was also developed to allow users to upload lung images and receive prediction results along with confidence scores and recommendations. The experimental results indicate that the proposed system can effectively assist in the early detection of lung cancer and has the potential to support AI-assisted medical diagnosis. Although further improvements are needed for real-world clinical applications, the developed system provides a promising foundation for future intelligent healthcare solutions.

9. References

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